

# INDIAN INSTITUTE OF TECHNOLOGY DELHI

New Delhi – 110016



## GIPEDI INTERNSHIP

Submitted in partial fulfillment of the requirements  
for the GIPEDI Internship Programme

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July 2026

# Certificate

This is to certify that the report entitled "**GIPEDI Internship Report**" submitted by **Mr. Ganesh Mahata (2022BB1197)**, Department of Biotechnology, Indian Institute of Technology Delhi, is a bonafide record of the work carried out during the GIPEDI Internship in the Department of Electrical Engineering, IIT Delhi, under my supervision during the period **15 May 2026 to 15 July 2026**.

The work presented in this report is original and fulfills the requirements of the internship programme.

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**Prof. Subrat Kar**  
Supervisor  
Department of Electrical Engineering  
Indian Institute of Technology Delhi

Date: \_\_\_\_\_

# Declaration

I hereby declare that this internship report entitled "**GIPEDI Internship Report**" is an original record of the work carried out by me during the internship conducted in the Department of Electrical Engineering, Indian Institute of Technology Delhi, under the guidance of **Prof. Subrat Kar** and the mentorship of **Mr. Chandan**.

The work presented in this report has not been submitted, either wholly or partially, for the award of any degree, diploma, or other academic qualification at this or any other institution.

**Ganesh Mahata**

2022BB1197

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# Acknowledgement

I would like to express my sincere gratitude to **Prof. Subrat Kar**, Department of Electrical Engineering, Indian Institute of Technology Delhi, for providing me with the opportunity to undertake this internship. His valuable guidance, encouragement, and insightful suggestions greatly contributed to the successful completion of this work.

I am especially thankful to my mentor, **Mr. Chandan**, for his continuous support, technical guidance, and constructive feedback throughout the internship. His practical knowledge and willingness to help enabled me to gain a deeper understanding of embedded systems and Arduino-based hardware development.

I also thank the faculty members, laboratory staff, and the Department of Electrical Engineering for providing the necessary facilities and a motivating environment to complete this internship successfully.

Finally, I express my heartfelt gratitude to my family and friends for their constant encouragement and support throughout this internship.

**Ganesh Mahata**

# Abstract

This report presents the work carried out during the GIPEDI Internship at the Department of Electrical Engineering, Indian Institute of Technology Delhi. The internship focused on embedded systems, sensor interfacing, and printed circuit board (PCB) design through two major engineering projects.

The first project involved the development of a load cell-based smart weighing system using a 1 kg load cell, HX711 amplifier, ESP32 development board, and SH1106 OLED display. A mechanical support frame was fabricated, the hardware was assembled, and firmware was developed for real-time weight measurement. The system was calibrated and experimentally evaluated by measuring small changes in weight, demonstrating reliable and repeatable performance.

The second project involved the design of a low-cost white noise generator PCB using KiCad. The work included schematic capture, footprint assignment, PCB layout design, design verification using ERC and DRC, 3D visualization, and generation of manufacturing outputs, including Gerber, NC drill, and Bill of Materials (BOM) files. PCB fabrication and hardware testing were outside the scope of the internship.

Overall, the internship provided practical experience in embedded system development, sensor interfacing, PCB design, hardware integration, firmware development, testing, and engineering documentation, thereby strengthening the foundation for future research and engineering applications.

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# Load Cell Based Smart Weighing System Using ESP32 and HX711

## 1.1 Introduction

Accurate weight measurement plays an important role in industrial automation, laboratories, healthcare, and smart manufacturing. In this project, a compact digital weighing system was developed using a **1 kg load cell**, an **HX711 precision amplifier**, an **ESP32 Dev Module**, and a **SH1106 OLED display** to perform real-time weight measurement. The primary objective was to develop the weighing system and experimentally evaluate the sensitivity of a low-capacity load cell.

A strain-gauge load cell generates only a small differential voltage proportional to the applied load, which cannot be measured directly by the ESP32. Therefore, the **HX711 24-bit amplifier and Analog-to-Digital Converter (ADC)** was used to amplify and digitize the sensor output. The ESP32 then applies calibration and tare correction, calculates the measured weight, and continuously displays the result on the OLED display.

To evaluate the measurement capability of the developed system, a practical sensitivity experiment was performed by gradually removing approximately **1 mL** of water from a container at regular intervals. Since the density of water is approximately **1 g/mL**, each removal produced a weight reduction of nearly **1 g**. The measured values were recorded after the display stabilized and compared with the expected weight to assess the sensitivity, stability, and repeatability of the system.

The developed prototype integrates the load cell, HX711 amplifier, ESP32, and OLED display into a compact weighing platform supported by a custom mechanical frame. The project provided practical experience in sensor interfacing, embedded programming, calibration, hardware integration, and experimental evaluation while demonstrating the suitability of the system for laboratory measurements and future IoT-based monitoring applications.

## 1.2 Objectives

The objectives of this project are as follows.

- To assemble a smart weighing system using a load cell, HX711 module, ESP32, and OLED display.
- To fabricate a mechanical frame for mounting the load cell and weighing platform.

- To interface the load cell with the ESP32 using the HX711 amplifier.
- To develop firmware for weight measurement and OLED display.
- To calibrate the weighing system for accurate measurements.
- To evaluate the performance of the developed system through experimental testing.

### 1.3 System Overview

The proposed smart weighing system integrates a load cell, HX711 amplifier module, ESP32 development board, SH1106 OLED display, and a custom mechanical support frame to perform real-time weight measurement. The system converts the mechanical force applied to the weighing platform into an electrical signal, which is amplified, digitized, processed, and displayed as weight.

Figure 1.1 illustrates the overall architecture of the developed system. When a load is placed on the weighing platform, the load cell produces a small differential voltage proportional to the applied force. The HX711 module amplifies this signal and converts it into digital data before transmitting it to the ESP32 through a serial interface. The ESP32 processes the received data using a calibration factor to determine the corresponding weight and displays the measured value on the SH1106 OLED display in real time.

A custom mechanical frame was fabricated to securely mount the load cell and weighing platform, ensuring stable operation and proper force transfer during measurements.

Load Cell → HX711 → ESP32 → OLED

**Figure 1.1:** Flow diagram of the proposed load cell-based smart weighing system.

The signal flow begins with the applied load and continues through the load cell, HX711 amplifier, ESP32 microcontroller, and OLED display, enabling accurate real-time weight measurement.

### 1.4 Hardware Description

The smart weighing system consists of four primary hardware components: a 1 kg single-point load cell, an HX711 load cell amplifier module, an ESP32 Dev Module, and a SH1106 OLED display. Each component performs a specific function in the overall measurement process. The following subsections briefly describe the role of each hardware component used in the project.

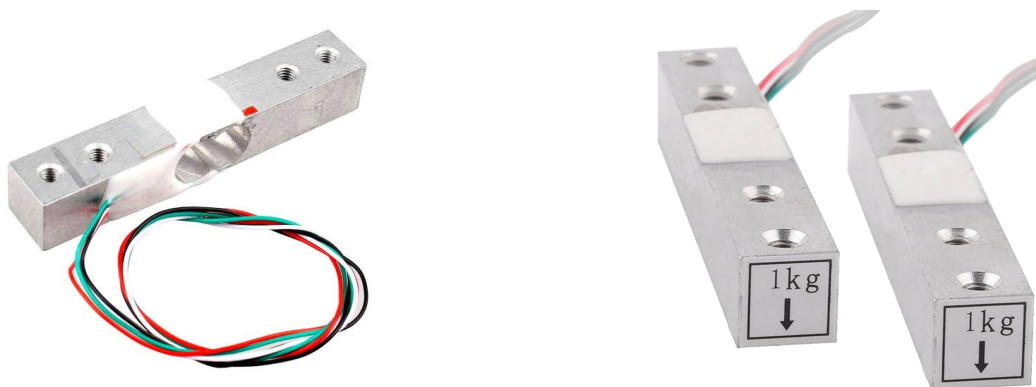
### 1.4.1 1 kg Load Cell

The load cell is the primary sensing element of the weighing system. It converts the mechanical force produced by the applied load into a small electrical signal.

A single-point strain-gauge load cell having a maximum capacity of 1 kg was selected for this project. This type of load cell is widely used in compact electronic weighing scales because of its compact size, good linearity, and high sensitivity. [1]

The selection of a 1 kg load cell was based on the objective of measuring relatively small changes in weight with good resolution. Since the maximum measurement range is limited to one kilogram, the sensor is capable of detecting smaller variations in applied load than higher-capacity load cells.

Figure 1.2 shows the load cell used during the development of the weighing system.



**Figure 1.2:** 1 kg Single-Point Load Cell (Front and Back View)

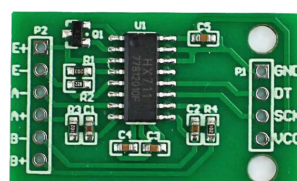
### 1.4.2 HX711 Load Cell Amplifier Module

The electrical signal generated by the load cell is extremely small and cannot be measured directly by the ESP32. Therefore, an HX711 precision amplifier module was used between the load cell and the microcontroller.

The HX711 amplifies the low-level signal produced by the load cell and converts it into a high-resolution digital value that can be processed by the ESP32. It is specifically designed for weighing applications and provides excellent measurement stability and resolution. [2]

The module communicates with the ESP32 using only two digital pins, making hardware interfacing simple while maintaining high measurement accuracy.

Figure 1.3 shows the HX711 module used in this project.



**Figure 1.3:** HX711 Load Cell Amplifier Module

### 1.4.3 ESP32 Dev Module

The ESP32 Dev Module acts as the central processing unit of the weighing system. It receives the digital data from the HX711 module, performs calibration and tare correction, calculates the actual weight, and updates the OLED display continuously.

The ESP32 was selected instead of a conventional microcontroller because it offers higher processing capability, larger memory, built-in Wi-Fi and Bluetooth connectivity, and sufficient GPIO pins for future expansion. [3]

Although wireless communication was not implemented in the final system, the availability of integrated Wi-Fi provides flexibility for future Internet of Things (IoT) applications such as remote monitoring and cloud-based data logging.

Figure 1.4 shows the ESP32 development board used during this project.

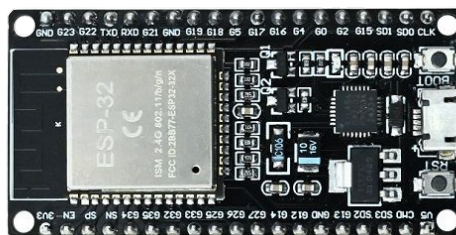


Figure 1.4: ESP32 Dev Module

### 1.4.4 SH1106 OLED Display

A 1.3-inch SH1106 OLED display was used to provide a real-time visual indication of the measured weight.

After every measurement cycle, the ESP32 updates the display with the latest weight value. During the calibration and sensitivity experiments, all observations were recorded only after the displayed value became stable.

The OLED display offers good contrast, low power consumption, and excellent readability, making it suitable for standalone operation without requiring continuous observation of the Serial Monitor. [4]

Figure 1.5 shows the OLED display used in the developed weighing system.



Figure 1.5: SH1106 OLED Display

## 1.5 Hardware Connections and Circuit Implementation

After selecting the required hardware components, the hardware prototype was assembled by interfacing the load cell with the HX711 amplifier module, the ESP32 Dev Module, and the SH1106 OLED display. Careful wiring of each component was necessary to ensure reliable communication and accurate weight measurement.

The complete hardware implementation consists of three major interfaces:

1. Load Cell to HX711 Module
2. HX711 Module to ESP32 Dev Module
3. SH1106 OLED Display to ESP32 Dev Module

### 1.5.1 Hardware Components

Table 1.1 lists the hardware used to develop the weighing system.

**Table 1.1:** Hardware Components Used

| Component                         | Quantity    |
|-----------------------------------|-------------|
| ESP32 Dev Module                  | 1           |
| 1 kg Single-Point Load Cell       | 1           |
| HX711 Load Cell Amplifier Module  | 1           |
| 1.3-inch SH1106 OLED Display      | 1           |
| USB Cable                         | 1           |
| Connecting Wires                  | As Required |
| Breadboard (Testing Stage)        | 1           |
| Laptop / Computer                 | 1           |
| Commercial Digital Weighing Scale | 1           |

### 1.5.2 Load Cell to HX711 Connections

The load cell is connected directly to the HX711 amplifier through four wires. These wires provide power to the Wheatstone bridge inside the load cell and transfer the differential output signal to the amplifier.

**Table 1.2:** Load Cell Connections

| Wire Colour | HX711 Pin | Purpose                      |
|-------------|-----------|------------------------------|
| Red         | E+        | Positive excitation voltage  |
| Black       | E-        | Ground / Negative excitation |
| Green       | A+        | Positive signal output       |
| White       | A-        | Negative signal output       |

### 1.5.3 HX711 to ESP32 Connections

The HX711 communicates digitally with the ESP32 using only two GPIO pins. Power is supplied directly from the ESP32 development board.

**Table 1.3:** HX711 to ESP32 Connections

| HX711 Pin | ESP32 Pin |
|-----------|-----------|
| VCC       | 3.3V      |
| GND       | GND       |
| DT (DOUT) | GPIO 32   |
| SCK       | GPIO 26   |

### 1.5.4 SH1106 OLED to ESP32 Connections

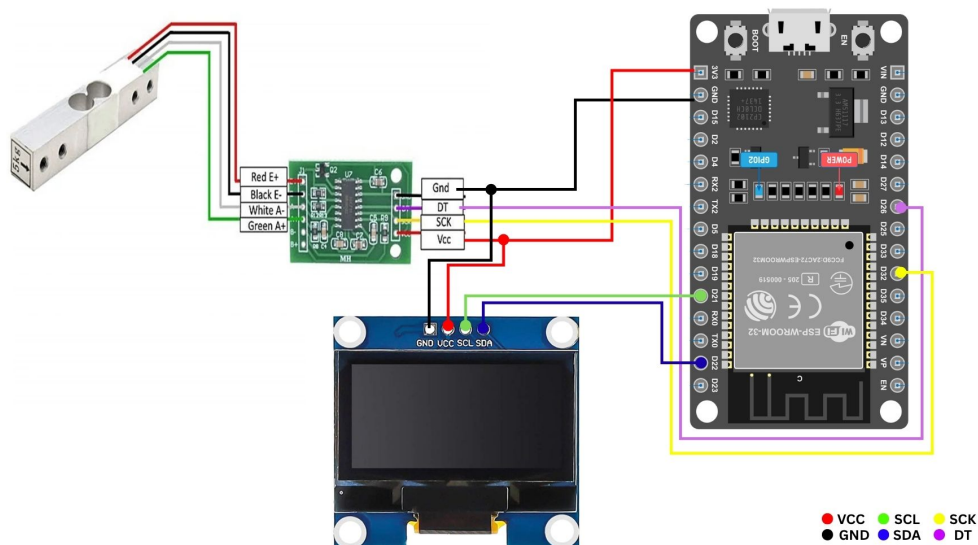
The SH1106 OLED display communicates with the ESP32 through the I<sup>2</sup>C interface.

**Table 1.4:** SH1106 OLED Connections

| OLED Pin | ESP32 Pin |
|----------|-----------|
| VCC      | 3.3V      |
| GND      | GND       |
| SDA      | GPIO 21   |
| SCL      | GPIO 22   |

### 1.5.5 Complete Circuit Diagram

Figure 1.6 illustrates the complete circuit diagram of the developed weighing system.

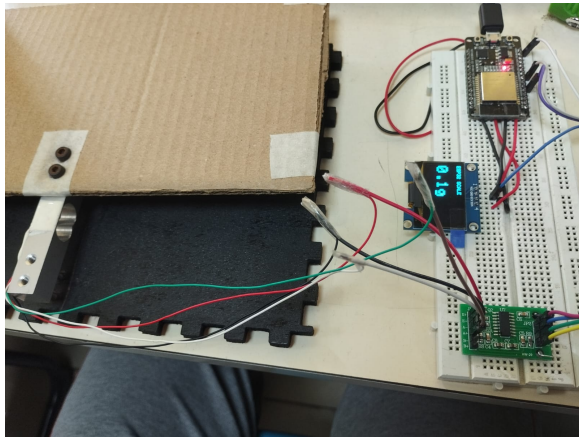


**Figure 1.6:** Complete Circuit Diagram of the Developed Weighing System

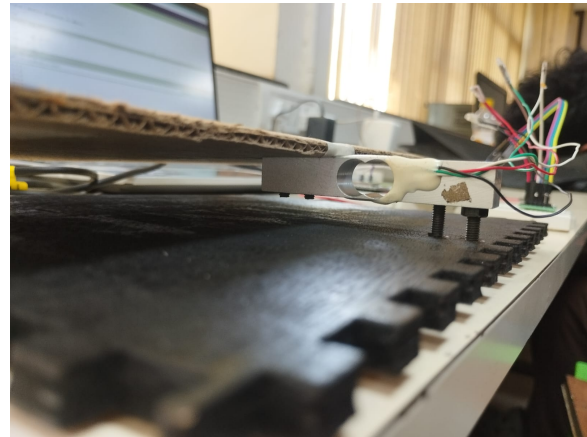
### 1.5.6 Experimental Hardware Setup

After completing the wiring, the entire system was assembled on the laboratory workbench. The load cell was securely mounted to minimize mechanical vibration during measurements, while the HX711 module and ESP32 were connected using jumper wires. The SH1106 OLED display was positioned for easy observation of the measured weight during calibration and sensitivity testing.

Figure 1.9 shows the actual hardware setup developed during the internship.



**Figure 1.7:** Complete Experimental Setup



**Figure 1.8:** Mounted Load Cell and Platform

**Figure 1.9:** Hardware Setup Used During Experimental Evaluation

## 1.6 Software Implementation

Firmware for the smart weighing system was written using the Arduino IDE with the ESP32 Dev Module as the target development board. The program is responsible for acquiring data from the HX711 module, processing the measured values, applying calibration, and displaying the final weight on the SH1106 OLED display.

During operation, the ESP32 continuously reads the digital output of the HX711 module, calculates the corresponding weight, and refreshes the OLED display in real time. To improve measurement stability, multiple sensor readings are averaged before displaying the final value.

### 1.6.1 Software Tools Used

The software and libraries used during the project's development are listed in Table 1.5. [5, 6, 7, 8]

**Table 1.5:** Software Development Tools

| Software / Library      | Purpose                                    |
|-------------------------|--------------------------------------------|
| Arduino IDE             | Program development and uploading firmware |
| ESP32 Board Package     | ESP32 board support                        |
| HX711 Library           | Reading weight from the load cell          |
| Adafruit SH110X Library | OLED display interface                     |
| Adafruit GFX Library    | Graphics and text display                  |
| Serial Monitor          | Calibration and debugging                  |

### 1.6.2 Calibration of the Load Cell

The load cell does not directly provide weight in grams. Instead, the HX711 module produces a raw digital output that is proportional to the applied load. Therefore, calibration is required to establish the relationship between the raw sensor output and the actual weight.

The calibration process was performed experimentally using a commercial digital weighing scale as the reference instrument. Initially, the raw output of the load cell was recorded without placing any object on the weighing platform. To reduce the effect of measurement noise, the average of multiple raw readings was considered as the zero-load reference value.

A known weight was then measured using the digital weighing scale and placed on the load cell platform. After allowing the readings to stabilize, the average of multiple raw readings was recorded again. The calibration factor was estimated using the relationship

$$\text{Calibration Factor} = \frac{\text{Signal with Test Weight} - \text{Signal at Zero}}{\text{Known Test Weight}} \quad (1.1)$$

where

- Signal at Zero is the average raw value measured with no applied load.
- Signal with Test Weight is the average raw value measured after placing the known weight.
- Known Test Weight is the reference weight measured using the commercial digital weighing scale.

### 1.6.3 Program Workflow

The calculated value provided an initial estimate of the calibration factor. The weight displayed on the Serial Monitor was compared with the reference weight. If the measured value differed from the reference, the calibration factor was adjusted, and the experiment was repeated until both values agreed closely.

Once the appropriate calibration factor was obtained, it was incorporated into the weighing system's final firmware.

The firmware follows a simple sequence of operations during execution.

---

**Algorithm 1** Firmware Execution

---

```
1: function SETUP
2:   Initialize ESP32
3:   Initialize HX711 module
4:   Initialize SH1106 OLED display
5:   Load calibration factor
6:   Perform tare operation
7: end function
8: function LOOP
9:   Read weight data from the HX711 module
10:  Average multiple sensor readings
11:  Calculate the weight
12:  Display the measured weight on the OLED display
13:  Send the measured weight to the Serial Monitor
14: end function
```

---

### 1.6.4 Weight Measurement

The HX711 library is used to obtain the weight measurements from the load cell. Instead of using a single sensor reading, the ESP32 averages twenty consecutive readings before calculating the displayed weight. This averaging technique reduces random fluctuations and provides a more stable measurement.

After processing the measured data, the calculated weight is displayed in grams on the SH1106 OLED.

### 1.6.5 Program Source Code

The complete source code is available in the GitHub repository, and the repository link is provided in the footer of this report.

## 1.7 Testing Methodology and Experimental Results

After completing the hardware assembly, software implementation, and calibration of the weighing system, experiments were conducted to evaluate its sensitivity and measurement accuracy.

A container containing approximately **100 g** of water was first weighed using a commercial digital weighing scale and then placed on the load cell platform. This initial weight served as the reference value throughout the experiment.

To investigate the minimum detectable change in weight, approximately **1 mL** of water was removed from the container using a syringe at regular intervals of about **15–20 minutes**. Since the density of water is approximately **1 g/mL**, each removal produced a weight reduction of nearly **1 g**.

After each removal, the weighing system was allowed to stabilize before recording the displayed weight from the SH1106 OLED display. Waiting for a stable reading minimized the influence of temporary fluctuations caused by mechanical vibrations and sensor settling.

The measured weight displayed by the developed system was then compared with the expected weight after each 1 mL reduction. This procedure was repeated several times to evaluate the sensitivity, accuracy, and repeatability of the developed weighing system.

The complete experimental procedure is summarized below:

1. Measure approximately 100 g of water using a commercial digital weighing scale.
2. Place the container on the calibrated load cell platform.
3. Wait until the displayed weight becomes stable and record the initial reading.
4. Remove approximately 1 mL of water using a syringe.
5. Wait again until the displayed weight stabilizes.
6. Record the new weight displayed on the OLED.
7. Compare the measured value with the expected weight.
8. Repeat the procedure several times to evaluate the sensitivity and accuracy of the system.

## 1.8 Observation Table

After completing the calibration process, The prototype weighing system was evaluated to determine its sensitivity and measurement accuracy. A container containing approximately 100 g of water was placed on the load cell platform. Subsequently, approximately 1 mL of water was removed using a syringe at regular intervals of 15 & 20 minutes.

Since the density of water is approximately 1 g/mL, each removal resulted in an expected decrease of nearly 1 g in the total weight. After every removal, the displayed weight on the SH1106 OLED was allowed to stabilize before recording the observation. After every removal, the system was allowed to stabilize before recording the displayed weight from the SH1106 OLED display.

The observed readings are presented in Table 1.6 & Table 1.7.

**Table 1.6:** Data Recorded Every 15 Minutes

| S.No. | Elapsed (min) | Water Removed (mL) | Measured Weight (g) | Rounded Weight (g) |
|-------|---------------|--------------------|---------------------|--------------------|
| 1     | 0             | 0                  | 100.02              | 100                |
| 2     | 15            | 1                  | 98.94               | 99                 |
| 3     | 30            | 1                  | 97.33               | 97                 |
| 4     | 45            | 1                  | 94.91               | 95                 |
| 5     | 60            | 1                  | 92.72               | 93                 |
| 6     | 75            | 1                  | 90.97               | 91                 |
| 7     | 90            | 1                  | 89.31               | 89                 |

**Table 1.7:** Data Recorded Every 20 Minutes

| S.No. | Elapsed Time<br>(min) | Water Removed<br>(mL) | Measured Weight<br>(g) | Rounded Weight<br>(g) |
|-------|-----------------------|-----------------------|------------------------|-----------------------|
| 1     | 0                     | 0                     | 100.28                 | 100                   |
| 2     | 20                    | 1                     | 98.96                  | 99                    |
| 3     | 40                    | 1                     | 97.58                  | 98                    |
| 4     | 60                    | 1                     | 95.72                  | 96                    |
| 5     | 80                    | 1                     | 93.84                  | 94                    |
| 6     | 100                   | 1                     | 92.72                  | 93                    |
| 7     | 120                   | 1                     | 90.74                  | 91                    |
| 8     | 140                   | 1                     | 89.09                  | 89                    |
| 9     | 160                   | 1                     | 87.78                  | 88                    |
| 10    | 180                   | 1                     | 85.88                  | 86                    |
| 11    | 200                   | 1                     | 83.95                  | 84                    |
| 12    | 220                   | 1                     | 82.15                  | 82                    |
| 13    | 240                   | 1                     | 81.27                  | 81                    |
| 14    | 260                   | 1                     | 79.46                  | 79                    |
| 15    | 280                   | 1                     | 77.75                  | 78                    |
| 16    | 300                   | 1                     | 76.25                  | 76                    |
| 17    | 320                   | 1                     | 75.49                  | 75                    |
| 18    | 340                   | 1                     | 73.92                  | 74                    |

The data presented in Table 1.6 & Table 1.7 demonstrate that the developed weighing system was capable of detecting small reductions in weight produced by the removal of approximately 1 mL of water during each observation. The measured values closely followed the expected trend throughout the experiment.

Minor deviations between the expected and measured weights were observed due to sensor noise, mechanical vibrations, and environmental factors. However, the absolute error remained within an acceptable range for the entire measurement period. Recording measurements only after the OLED display value became stable and averaging multiple sensor readings significantly improved the system's repeatability and stability.

These observations confirm that the developed weighing system is sufficiently sensitive to detect weight variations of approximately 1 g, demonstrating the effectiveness of the calibration procedure and the overall performance of the load cell, HX711 module, and ESP32-based measurement system.

## 1.9 Results and Discussion

The sensitivity evaluation demonstrated that the weighing system reliably detected small variations in weight throughout the experiment. A consistent decrease in measured weight was observed after each withdrawal of approximately **1 mL** of water, confirming that the system responded appropriately to incremental changes in load. This behavior indicates

that the integration of the **load cell**, the **HX711 amplifier** and the **ESP32 micro-controller** provided an effective and reliable weight measurement.

The recorded measurements exhibited a clear decreasing trend that closely matched the expected reduction in water mass. Although minor fluctuations were present between successive readings, these variations remained small and did not affect the overall consistency of the results. Such fluctuations are expected in practical sensing systems due to *electrical noise*, *environmental disturbances*, and the high sensitivity of strain-gauge-based load cells.

Measurement stability was significantly improved by averaging **20 consecutive sensor readings** before displaying the final value. This averaging technique effectively minimized random noise and reduced the influence of small mechanical vibrations, resulting in smoother and more repeatable measurements. In addition, data were recorded only after the OLED display reached a stable value, ensuring consistency across all observations.

The experiment also highlighted certain practical limitations associated with manual operation. Accurately withdrawing exactly **1 mL** of water using a syringe during each trial was challenging, leading to small variations in the actual quantity removed. Consequently, slight differences between the expected and measured weights were observed. These deviations are primarily attributed to **human handling errors** rather than limitations of the sensing hardware or measurement algorithm.

Overall, the sensitivity test confirmed that the developed weighing system could successfully detect weight changes of approximately **1 g**, corresponding to nearly **1 mL** of water. The experimental results demonstrate that the system provides **stable**, **repeatable**, and **sufficiently accurate** measurements within the operating range of the **1 kg load cell**. These findings validate the effectiveness of the proposed design for applications requiring **real-time monitoring of small weight variations** and demonstrate its potential for deployment in *laboratory experiments*, *industrial monitoring*, and *IoT-based sensing applications*.

## 1.10 Future Scope

Although the proposed weighing system successfully measures weight with good sensitivity and stability, several improvements can further enhance its functionality and practical applications.

- **Automatic Data Logging:** The current system displays the measured weight on the SH1106 OLED display. In future, the ESP32 can automatically transmit every measured reading to a cloud database without requiring manual recording.
- **Backend Database Integration:** The weighing system can be integrated with a backend server using technologies such as Node.js and PostgreSQL. This would enable secure storage of historical weight measurements along with timestamps for future analysis.
- **Real-Time Web Dashboard:** The stored data can be processed and visualized through a web-based dashboard. Platforms such as Grafana can be integrated

with the database to display live weight measurements, historical trends, statistical summaries, and interactive graphs.

- **Remote Monitoring:** Since the ESP32 provides built-in Wi-Fi connectivity, the developed system can be extended for remote monitoring, allowing users to observe weight measurements from any location through a web application.
- **Alert and Notification System:** Automatic notifications can be generated whenever the measured weight exceeds or falls below predefined threshold values. Such alerts can be delivered through email, SMS, or mobile applications.
- **Industrial and IoT Applications:** The developed system can be integrated into industrial automation, smart inventory management, laboratory monitoring, and Internet of Things (IoT) applications where continuous weight monitoring and automatic data collection are required.

# PCB Design of a Low-Cost White Noise Generator

## 2.1 Introduction

Printed Circuit Boards (PCBs) are widely used in modern electronic systems because they provide a compact, reliable, and permanent platform for electronic circuits. Compared to breadboard implementations, PCBs improve circuit stability, reduce unwanted wiring, and make the circuit easier to manufacture and maintain.

During this internship, a low-cost White Noise Generator PCB was designed using **KiCad**, an open-source Electronic Design Automation (EDA) software. The objective of this work was to convert an existing white noise generator circuit into a compact and manufacturable PCB design by following standard PCB design practices.

The work included schematic capture, component selection, footprint assignment, PCB layout design, component placement, copper trace routing, design verification, and 3D visualization of the final board. Since the scope of this internship was limited to PCB design, fabrication and hardware testing were not performed.

The final PCB layout serves as a ready-to-manufacture prototype and provides the foundation for future fabrication, assembly, and experimental evaluation.

## 2.2 Objectives

The main objectives of this work are listed below.

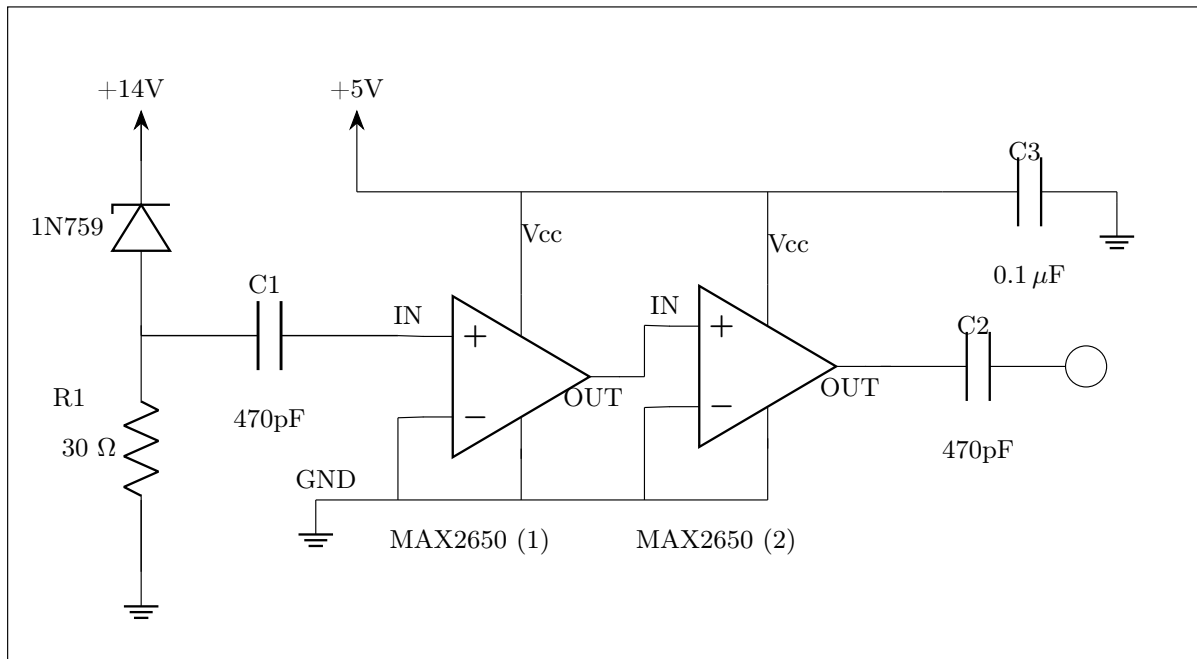
- Study the working principle of a low-cost white noise generator.
- Recreate the circuit schematic using KiCad.
- Select suitable electronic components and PCB footprints.
- Design a compact and manufacturable PCB layout.
- Perform proper component placement and copper trace routing.
- Verify the PCB design using Electrical Rule Check (ERC) and Design Rule Check (DRC).
- Generate a 3D model of the PCB for visual inspection.
- Prepare the PCB design for future fabrication.

## 2.3 System Overview

The White Noise Generator is an analog electronic circuit designed to generate a broadband random noise signal. Such signals are commonly used in communication systems, electronic testing, audio applications, and laboratory experiments.

In this project, the complete circuit was designed in KiCad by first creating the schematic and then converting it into a PCB layout. Appropriate component footprints were assigned, the PCB was routed, and the design was verified before generating a 3D model of the board.

Figure 2.1 presents the reference circuit schematic used for designing the PCB. The original circuit was adapted from Reference [9].



**Figure 2.1:** Circuit schematic of the White Noise Generator using two MAX2650 low-noise amplifiers.

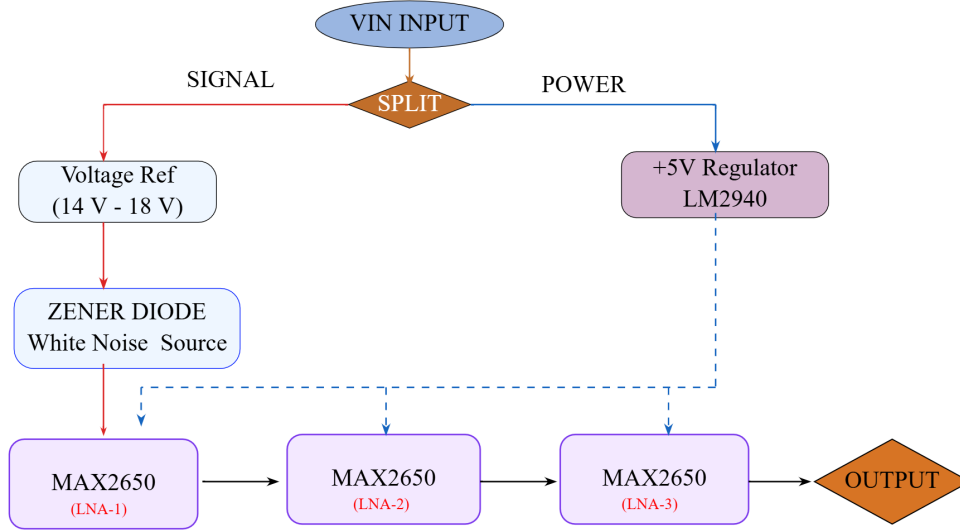
## 2.4 Working Principle of the White Noise Generator

White noise is a random electrical signal that contains a wide range of frequency components with nearly uniform power over a specified frequency range. Due to its broadband nature, white noise is commonly used in communication systems, electronic testing, filter characterization, and audio applications.

The white noise generator designed in this project is based on the avalanche breakdown phenomenon of a reverse-biased Zener diode. When the reverse voltage applied across the Zener diode exceeds its breakdown voltage, random movement of charge carriers produces a very small electrical noise signal. Although this signal has the characteristics of white noise, its amplitude is too low for practical use.

To obtain a usable output, the generated noise signal is amplified using three cascaded

MAX2650 low-noise amplifier (LNA) stages. These amplifiers increase the signal amplitude while maintaining a wide operating bandwidth. A regulated +5 V power supply, provided by the LM2940 voltage regulator, ensures stable operation of the active components. In addition, an adjustable reference circuit using the AZ9431BQ shunt regulator and TS53 trimmer allows the operating point of the Zener diode to be fine-tuned.



**Figure 2.2:** Functional Block Diagram of the White Noise Generator

## 2.5 Circuit Schematic Design

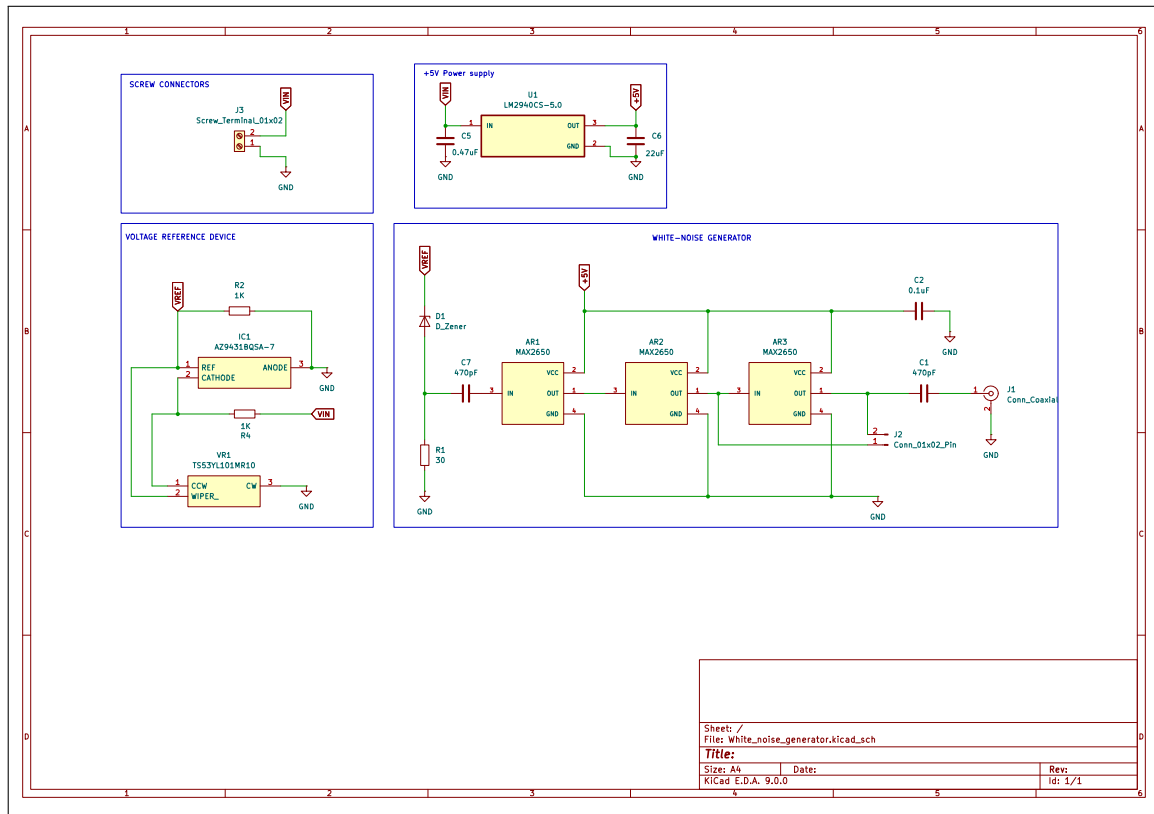
The circuit schematic was developed using the KiCad Electronic Design Automation (EDA) software. The objective of the schematic design was to accurately represent all electrical connections of the white noise generator before proceeding to PCB layout design.

The schematic consists of four main functional blocks: the power supply, bias and reference circuit, white noise generation stage, and amplifier stage. Each block performs a specific function and together they generate and amplify the white noise signal.

The design process began by placing the required electronic components in the KiCad schematic editor. Appropriate symbols were selected from the KiCad component libraries, and all electrical connections were completed according to the circuit design. Power symbols, ground connections, and net labels were added to improve readability and simplify circuit verification.

After completing the schematic, an Electrical Rule Check (ERC) was performed to identify missing connections, incorrect pin configurations, or other electrical errors. All reported issues were corrected before proceeding to the PCB layout stage.

The completed schematic of the White Noise Generator is shown in Figure 2.3.



**Figure 2.3:** Noise Generator Circuit - Schematic

## 2.5.1 Functional Blocks of the Circuit

The complete circuit can be divided into four functional blocks, each responsible for a specific operation.

### 1. Power Supply Block

The LM2940 voltage regulator converts the input supply into a stable +5 V output required by the active components of the circuit.

### 2. Bias and Reference Circuit

The AZ9431BQ precision shunt regulator and TS53 trimmer establish the operating voltage for the reverse-biased Zener diode, ensuring stable noise generation.

### 3. White Noise Generation Block

The reverse-biased Zener diode operates in avalanche breakdown mode and generates a low-level broadband white noise signal.

### 4. Amplifier Block

Three MAX2650 low-noise amplifier stages are connected in cascade to amplify the weak noise signal and produce a usable output.

## 2.6 PCB Design Methodology

After completing and verifying the circuit schematic, the next step was to design the Printed Circuit Board (PCB) using KiCad. The objective was to develop a compact, well-organized, and manufacturable PCB while following standard PCB design practices.

PCB development involved assigning footprints to all components, arranging them on the board, routing electrical connections, and verifying the completed layout before generating manufacturing files.

### 2.6.1 Footprint Assignment

Each component in the schematic was assigned an appropriate PCB footprint using the KiCad footprint libraries. The selected footprints matched the package dimensions specified in the respective component datasheets to ensure correct placement during PCB fabrication and assembly.

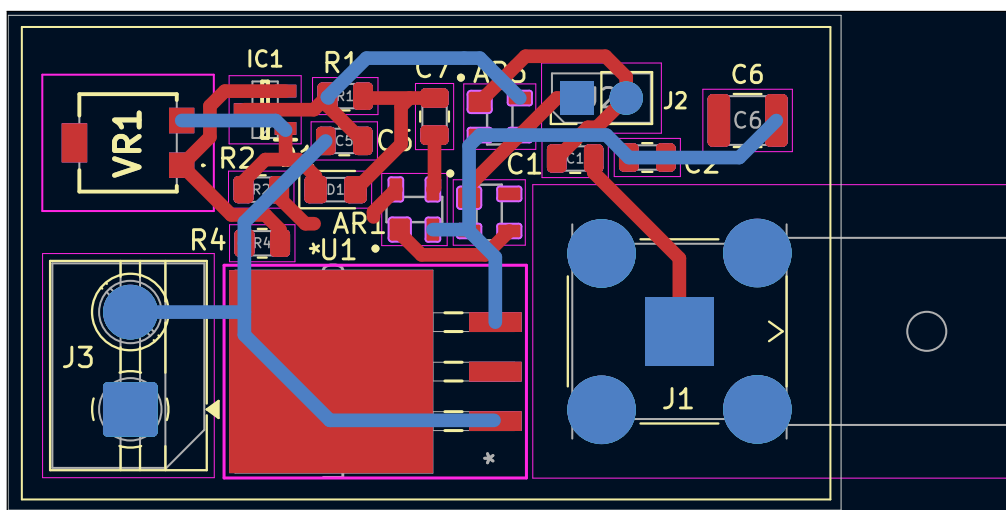
After assigning footprints, the footprint associations were verified to confirm that every schematic symbol was linked to its corresponding PCB package.

### 2.6.2 PCB Layout Design

The PCB layout was created by importing the completed schematic into the KiCad PCB Editor. Initially, all components appeared outside the board outline. A suitable board size was selected, and the components were arranged systematically to obtain a compact and organized layout.

The placement was planned to reduce unnecessary routing complexity while maintaining sufficient spacing between components for reliable manufacturing.

Figure 2.4 shows the completed PCB layout designed in KiCad.



**Figure 2.4:** Noise Generator- PCB Layout

### **2.6.3 Component Placement**

Proper component placement is an important step in PCB design because it directly affects routing quality, board size, and overall circuit performance.

Components performing similar functions were grouped together to simplify routing. The amplifier stages were placed sequentially to maintain the signal flow, while the power supply and biasing components were positioned close to the circuits they support. This arrangement reduced routing complexity and improved the overall organization of the PCB.

### **2.6.4 Copper Trace Routing**

After component placement, electrical connections were completed by routing copper traces between the components. Appropriate trace widths were selected according to the power and signal requirements of the circuit.

Short and direct routing paths were preferred wherever possible to minimize unnecessary trace length. The routing was completed while avoiding overlapping traces and maintaining adequate spacing between adjacent tracks.

### **2.6.5 Ground Plane**

A ground copper pour was added to the PCB to provide a common reference potential for the circuit. The ground plane also improves electrical connectivity and simplifies PCB routing by reducing the number of individual ground traces.

### **2.6.6 Design Verification**

The completed PCB layout was verified using the Design Rule Check (DRC) tool available in KiCad. The verification process checked for routing errors, clearance violations, overlapping tracks, and other manufacturing issues.

After correcting all identified issues, the PCB layout was considered ready for manufacturing file generation.

### **2.6.7 3D PCB Visualization**

KiCad provides a built-in 3D Viewer that allows the designed PCB to be inspected before fabrication. The 3D model was used to verify component placement, footprint orientation, and the overall appearance of the board.

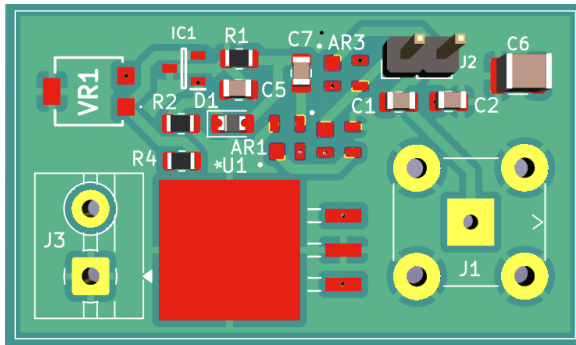


Figure 2.5: Top View

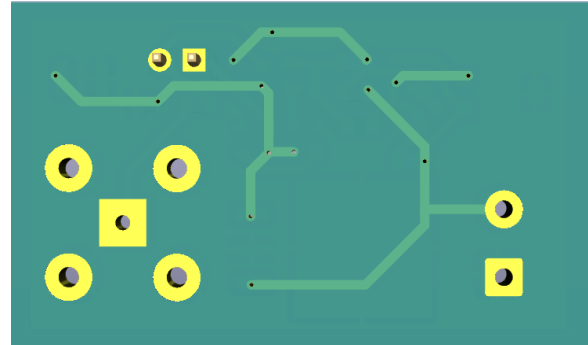


Figure 2.6: Bottom View

## 2.7 Manufacturing Outputs

After completing the PCB layout and verifying the design using the Design Rule Check (DRC), the manufacturing files required for PCB fabrication were generated in KiCad. These files prepare the design for future fabrication and assembly. As the primary objective of this internship was PCB design, fabrication and hardware testing were outside the scope of this work.

### 2.7.1 Gerber and Drill Files

The complete set of Gerber files and NC drill files was generated using KiCad. The Gerber files contain the information for the copper layers, solder mask, silkscreen, and board outline, while the drill files specify the locations and diameters of all drilled holes required during PCB manufacturing.

The generated manufacturing files are included in the project repository and can be directly used for PCB fabrication.

### 2.7.2 Bill of Materials (BOM)

A Bill of Materials (BOM) was automatically generated using KiCad after completing the PCB design. The BOM lists all components required for PCB assembly, including their reference designators, quantities, component values, and PCB footprints. The complete BOM, along with the Gerber and NC drill files, is available in the project repository.

**Table 2.1:** Bill of Materials (BOM) Generated Using KiCad

| Reference | Qty | Value             | Footprint              |
|-----------|-----|-------------------|------------------------|
| AR1–AR3   | 3   | MAX2650           | SOT190P228X119-4N      |
| C1, C7    | 2   | 470 pF            | 0805                   |
| C2        | 1   | 0.1 $\mu$ F       | 0805                   |
| C5        | 1   | 0.47 $\mu$ F      | 0805                   |
| C6        | 1   | 22 $\mu$ F        | 1210                   |
| D1        | 1   | 1N759 Zener Diode | D_0805                 |
| IC1       | 1   | AZ9431BQSA-7      | SOT96P240X110-3N       |
| J1        | 1   | BNC Connector     | BNC_WNG1               |
| J2        | 1   | 2-Pin Header      | Pin Header 2.54 mm     |
| J3        | 1   | Screw Terminal    | 5.08 mm Terminal Block |
| R1        | 1   | 30 $\Omega$       | 0805                   |
| R2, R4    | 2   | 1 k $\Omega$      | 0805                   |
| U1        | 1   | LM2940IMP-9.0     | TO254P1524X483-4N      |
| VR1       | 1   | TS53YL101MR10     | TS53                   |

## 2.8 Results and Discussion

The White Noise Generator PCB was successfully designed using KiCad by following the standard PCB design workflow. The schematic was recreated from the reference circuit, footprints were assigned to all components, and a compact PCB layout was developed.

The completed PCB was verified using Electrical Rule Check (ERC) and Design Rule Check (DRC) to identify and eliminate design errors. A ground plane was added, copper traces were routed, and the final layout was inspected using KiCad's 3D Viewer. Manufacturing outputs, including Gerber files, drill files, and the Bill of Materials (BOM), were generated successfully.

Although the PCB was not fabricated or experimentally tested during the internship, the completed design is suitable for future manufacturing and hardware implementation.

## 2.9 Future Scope

The work completed during this internship provides a foundation for future development of the White Noise Generator. The following activities can be carried out in future work:

- Fabrication of the designed PCB using the generated Gerber files.
- Assembly and soldering of all electronic components on the PCB.
- Experimental testing and performance evaluation of the circuit.

- Measurement of output characteristics using laboratory instruments such as an oscilloscope and spectrum analyzer.
- Optimization of the PCB layout to further reduce electrical noise and improve signal quality.
- Integration of the White Noise Generator into RF communication and electronic testing applications.

## 2.10 Conclusion

During this internship, the complete PCB design of a White Noise Generator was developed using KiCad. The work included schematic capture, footprint assignment, PCB layout design, component placement, copper trace routing, design verification, 3D visualization, and generation of manufacturing files.

The internship provided practical experience in professional PCB design techniques and the use of modern Electronic Design Automation (EDA) tools. The completed design serves as a manufacturing-ready PCB that can be fabricated and tested in future work.

# Overall Conclusion

The GIPEDI Internship at the Department of Electrical Engineering, Indian Institute of Technology Delhi provided valuable hands-on experience in embedded systems, electronic hardware development, Arduino programming, and PCB design. Throughout the internship, theoretical concepts learned during academic coursework were applied to practical engineering problems, strengthening both technical knowledge and problem-solving skills.

During the internship, several embedded system experiments and applications were studied and implemented using the Arduino platform. These included sensor interfacing, display modules, real-time clock applications, EEPROM data logging, motor control, waveform generation, weather monitoring, security systems, and other microcontroller-based experiments. These activities strengthened my understanding of embedded programming, digital and analog interfacing, communication protocols, and hardware debugging while providing a strong practical foundation in embedded system design.

The first major project focused on the development of a load cell-based smart weighing system using a 1 kg load cell, an HX711 precision amplifier, an ESP32 Dev Module, and a SH1106 OLED display. The system was successfully designed, assembled, calibrated, and experimentally evaluated. The sensitivity analysis demonstrated that the developed prototype could reliably detect weight variations of approximately 1 g while providing stable and repeatable measurements. This project provided practical experience in sensor interfacing, calibration techniques, embedded programming, hardware integration, and experimental data analysis.

The second major project involved the design of a low-cost White Noise Generator Printed Circuit Board (PCB) using KiCad. The complete workflow, including schematic capture, footprint assignment, component placement, copper trace routing, design verification, and generation of manufacturing outputs, was successfully completed by following standard PCB design practices. Although PCB fabrication and hardware testing were outside the scope of the internship, the final PCB design serves as a manufacturable prototype suitable for future implementation.

Overall, this internship significantly enhanced my practical skills in embedded system development, electronic circuit design, PCB layout design, debugging, technical documentation, and engineering problem solving. It also provided valuable exposure to professional engineering tools and technologies such as Arduino IDE, ESP32, KiCad, SimulIDE, sensors, communication modules, and modern PCB design methodologies. The knowledge and experience gained during this internship have established a strong foundation for future academic research, industrial projects, and the development of advanced embedded and Internet of Things (IoT) systems.

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